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Effect of 810-nm diode laser photobiomodulation on the reversal of maxillary and mandibular soft tissue local anaesthesia in children: a randomized split-mouth clinical trial

Ayça Kurt[1]✉, Elif Kibaroğlu[2], Hatice Yıldırım[3], Zeynep Gümrükçü[4] & Emre Balaban[5]

This randomized split-mouth clinical trial evaluated the effectiveness of 810-nm diode laser photobiomodulation therapy (PBMT) in accelerating reversal of maxillary and mandibular soft tissue local anaesthesia in children. Forty children aged 4–9 years requiring infiltration anesthesia in both arches were included in the study. Each participant received PBMT irradiation on one side and sham irradiation on the opposite side. Irradiation was performed 45 min after the application of local anesthesia. The primary outcome was the time required for the return of normal soft tissue sensation. Secondary outcomes included the incidence of postoperative soft tissue trauma. Recovery time was significantly shorter in the PBMT group than in the sham group ($p < 0.001$). At 20 and 30 min, recovery rates were significantly higher in the PBMT group ($p = 0.002$ and $p = 0.035$, respectively). Soft tissue trauma incidence was lower in the PBMT group ($p = 0.020$). Comparable recovery trends were observed in both maxillary and mandibular procedures. PBMT using an 810-nm diode laser appears to be a safe, non-invasive, and effective adjunct for reducing the duration of soft tissue local anaesthesia and associated trauma in paediatric dental patients. *This study was registered on ClinicalTrials.gov (Protocol ID: NCT06841120), 2025-21-01, retrospectively registered.

Keywords Photobiomodulation therapy, Diode laser, Local anaesthesia, Paediatric dentistry, Soft tissue anaesthesia

Effective pain control achieved through local anaesthesia is essential in paediatric dentistry, as it enables successful dental treatment and promotes positive cooperation between the child and the clinician[1]. However, the effects of local anaesthesia—particularly soft tissue numbness—may persist for several hours following dental procedures[2]. In children, prolonged numbness may result in unintentional biting, sucking, or chewing of the lips, cheeks, or tongue, leading to self-inflicted soft tissue trauma and post-operative discomfort[3,4].

Several approaches have been investigated to shorten the duration of soft tissue anaesthesia and reduce its associated complications[5–8]. Pharmacological reversal agents such as phentolamine mesylate have demonstrated effectiveness; however, their invasive nature and pharmacological profile may limit acceptance in paediatric populations[8]. Consequently, interest has increased in non-invasive, drug-free methods capable of accelerating anaesthetic recovery[6,9].

[1]Department of Paediatric Dentistry, Faculty of Dentistry, Recep Tayyip Erdogan University, Rize, Turkey. [2]Kocaeli Provincial Health Directorate, Darıca Oral and Dental Health Centre, Kocaeli, Turkey. [3]Büyükçekmece Oral and Dental Health Center, Istanbul, Turkey. [4]Department of Oral and Maxillofacial Surgery, Faculty of Dentistry, Recep Tayyip Erdogan University, Rize, Turkey. [5]Department of Oral and Maxillofacial Surgery, Faculty of Dentistry, Ankara Yildirim Beyazit University, Ankara, Turkey. ✉email: ayca.kurt@erdogan.edu.tr

Photobiomodulation therapy (PBMT) is a non-thermal and non-invasive method that stimulates biological responses at the cellular and tissue level[10–12]. PBMT has been shown to exert analgesic, anti-inflammatory, and regenerative effects through mechanisms such as enhanced mitochondrial activity, increased adenosine triphosphate production, and modulation of inflammatory mediators[13–15]. Importantly, PBMT has been associated with vasodilation and increased microcirculation, which may facilitate the accelerated clearance of local anaesthetic agents from soft tissues[9,16].

Diode lasers operating in the red and near-infrared spectrum (630–980 nm) are commonly used for PBMT due to their favourable tissue penetration and safety profile[14,17]. Experimental and clinical studies have demonstrated that near-infrared diode lasers can increase local blood flow through endothelium-dependent vasodilation, suggesting a potential role in shortening the duration of soft tissue anaesthesia[18,19]. Previous studies have demonstrated the beneficial effects of laser-based therapies in the management of paediatric oral soft tissue conditions, supporting the biological rationale and clinical applicability of PBMT in paediatric dentistry[14,15]. In paediatric dentistry, PBMT has been investigated for various applications[14,15], including pain reduction during local anaesthetic injection and reversal of soft tissue anaesthesia, primarily in the mandibular and maxillary regions[6,9,20].

Although previous studies have reported promising results regarding the use of PBMT to reverse mandibular and maxillary soft tissue anaesthesia in children, comparative studies performed simultaneously in both arches are, to our knowledge, limited[6,9,20].

Therefore, the present randomized split-mouth clinical trial aimed to assess the effectiveness of PBMT using an 810-nm diode laser in reducing the duration of soft tissue local anaesthesia in both the maxillary and mandibular arches of children aged 4–9 years. The null hypothesis was that PBMT would have no effect on the duration of soft tissue anaesthesia, whereas the alternative hypothesis proposed that PBMT would significantly shorten anaesthetic recovery time.

Materials and methods

Study design

This randomized, split-mouth, double-blind clinical trial was conducted at the Faculty of Dentistry, Recep Tayyip Erdogan University. The study design followed the Consolidated Standards of Reporting Trials (CONSORT) guidelines. Each participant served as their own control, receiving PBMT on one side and sham laser irradiation on the contralateral side in separate sessions.

Ethical approval and trial registration

Ethical approval and study permits were obtained from the Recep Tayyip Erdoğan University Invasive Clinical Research Ethics Committee and the Turkish Ministry of Health, Turkish Medicines and Medical Devices Agency for our study (Approval No: E-85521274-000-2076076, Version Code: V:04). For children aged 3–8 years participating in this study, a written informed consent form specific to this age range was obtained; for children aged 9 years and older, a separate written informed consent form appropriate to this age range was obtained; in addition, written informed consent forms were obtained from the parents of all patients. Our study was conducted in accordance with the ethical standards specified in the 1964 Helsinki Declaration and subsequent or similar ethical standards.

No significant changes unforeseen in the plan occurred after the study began. Therefore, there are no unforeseen or undetermined results or analyses.

Sample size calculation

The sample size was calculated based on the effect size reported by Seraj et al.[9], which demonstrated a Cohen's *d* of 2.35 for the reduction of soft tissue anaesthesia duration using PBMT. With a type I error rate (α) of 0.05 and a statistical power of 95%, a minimum of 24 sites (12 per group) was required. To account for potential dropouts and to allow subgroup analyses of maxillary and mandibular arches, a total of 40 children were included in the study (Fig. 1). Sample size calculation was performed using G*Power software (version 3.0.10; University of Kiel, Germany).

No interim analyses were obtained during the study, and no situations arose that would necessitate stopping the study.

Participants

Children aged 4–9 years who required local infiltration anaesthesia in both maxillary and mandibular dental arches were screened for eligibility.

Inclusion criteria

- Children whose parents or legal guardians provided written informed consent and who gave assent.
- Frankl behaviour rating scale score of 3 or 4.
- Ability to understand and respond to soft tissue sensation.
- Requirement for dental treatment under local infiltration anaesthesia in both dental arches.
- No history of allergy to local anaesthetic agents or sulphites.
- No analgesic medication use within 24 h prior to treatment.
- Absence of systemic conditions that could interfere with dental treatment or anaesthesia.

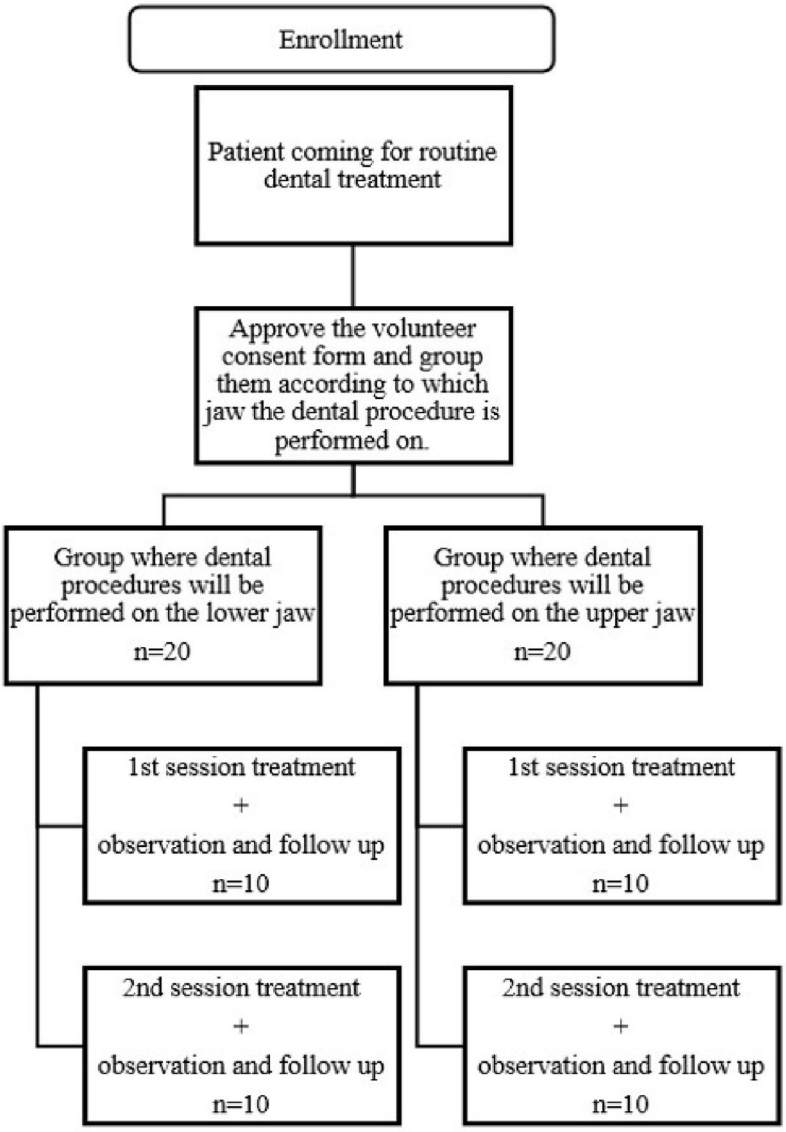


Fig. 1. Flowchart of the study.

Exclusion criteria

- Inadequate anaesthesia due to technical errors during injection.
- Inability to provide reliable sensory feedback.
- Systemic conditions contraindicating local anaesthesia.
- Need for urgent dental treatment due to acute pain or infection.

Randomization and blinding

The study was conducted over two treatment sessions. In one session, PBMT was applied following local anaesthesia, whereas in the other session, sham laser irradiation was administered. Allocation of laser and sham laser sessions and sides was determined by the participant drawing one of eight identical sealed envelopes prepared by an independent investigator.

The study employed a double-blind design: neither the participants nor the outcome assessor were aware of the allocation. Sham laser irradiation was performed using the same handpiece and application time, without laser emission.

Local anaesthesia administration

All local anaesthetic injections were performed by the same clinician to ensure consistency. A full cartridge containing 80 mg articaine hydrochloride with epinephrine (ULTRAVER D-S; 1:100,000) was administered for local infiltration anaesthesia. Anaesthetic effectiveness was confirmed using palpation and needle penetration tests prior to treatment.

Laser irradiation protocol (photobiomodulation therapy)

Photobiomodulation therapy (PBMT) was applied approximately 45 min after local anaesthetic administration. An 810-nm diode laser device (Fox; ARC Laser Gmbh, Nürnberg, Germany) was used. The laser was operated in continuous wave mode with a mean output power of 0.4 W and a duty cycle of 100%. Irradiation was delivered through a therapy handpiece with a spot size of 0.38 cm^2. PBMT was applied at six standardized points over the anaesthetized soft tissue area using a non-contact technique. Each point received 12 s of irradiation, corresponding to 4.8 J energy per point and a fluence of 12.6 J/cm^2. The irradiance was 1.05 W/cm^2, and the total delivered energy per session was 28.8 J. All laser parameters were selected according to previously published PBMT protocols and manufacturer recommendations (Table 1).

Outcome assessment

The primary outcome was the time required for the return of normal soft tissue sensation, assessed using palpation and needle penetration tests at 10-minute intervals following laser or sham irradiation. Sensory responses were categorized as numbness, tingling sensation, or normal sensation. The secondary outcome was the incidence of soft tissue trauma, assessed 24 h post-treatment through caregiver reports and photographic documentation when available.

A schematic overview of the study design, the PBMT implementation protocol, and the results evaluation is shown in Fig. 2.

Statistical analysis

Statistical analysis was performed using Jamovi software (version 2.3.21). Data normality was assessed using the Kolmogorov–Smirnov test. Continuous variables were analysed using paired sample *t*-tests or Student's *t*-tests, as appropriate. Categorical variables were compared using the McNemar paired sample test. Statistical significance was set at $p < 0.05$. The analysis did not include any missing data.

Results

A total of 40 paediatric patients (24 males, 16 females) with a mean age of 7.3 ± 1.4 years (range: 4–9 years) completed the study. All participants were included in the final analysis, and no dropouts or protocol deviations were recorded (Table 2).

The mean recovery time for return of normal soft tissue sensation was significantly shorter in the PBMT group compared with the sham group (163.0 ± 53.0 min vs. 206.9 ± 83.8 min, $p < 0.001$; Table 3). At 20 min, return of normal sensation was observed in 45% of participants in the PBMT group and 10% in the sham group ($p = 0.002$; Table 3). Similarly, at 30 min, return of normal sensation rates were significantly higher in the PBMT group (72.5% vs. 35%, $p = 0.035$; Table 3). The cumulative proportion of participants with return of normal soft tissue sensation remained higher in the PBMT group than in the sham group throughout the observation period (10–30 min). The corresponding 95% confidence intervals for each time point are presented in Fig. 3. **Soft tissue trauma incidence was lower in the PBMT group than in the sham group (5% vs. 22.5%**, $p = 0.020$).

Subgroup analyses demonstrated a significantly shorter anaesthesia duration in the laser group across both maxillary and mandibular arches ($p < 0.05$). Age-based analysis revealed significantly faster recovery in both age groups (≤7 years and >7 years) following PBMT ($p < 0.05$). A statistically significant gender-related difference was observed only among female participants, with shorter anaesthesia duration in the laser-treated sites ($p = 0.008$) (Table 4). Since no harm or adverse events were observed in the groups, this information was not included in the results.

Parameter	Value
Wavelength	810 nm
Laser type	Diode laser
Emission mode	Continuous wave (CW)
Mean output power	0.4 W
Peak power	Not applicable (CW mode)
Duty cycle	100%
Spot size at tissue	0.38 cm^2
Irradiation time per point	12 s
Irradiance (power density)	1.05 W/cm^2
Energy per point	4.8 J
Fluence per point	12.6 J/cm^2
Number of irradiation points	6
Total delivered energy	28.8 J
Application mode	Non-contact, stationary

Table 1. Laser parameters used for photobiomodulation therapy.

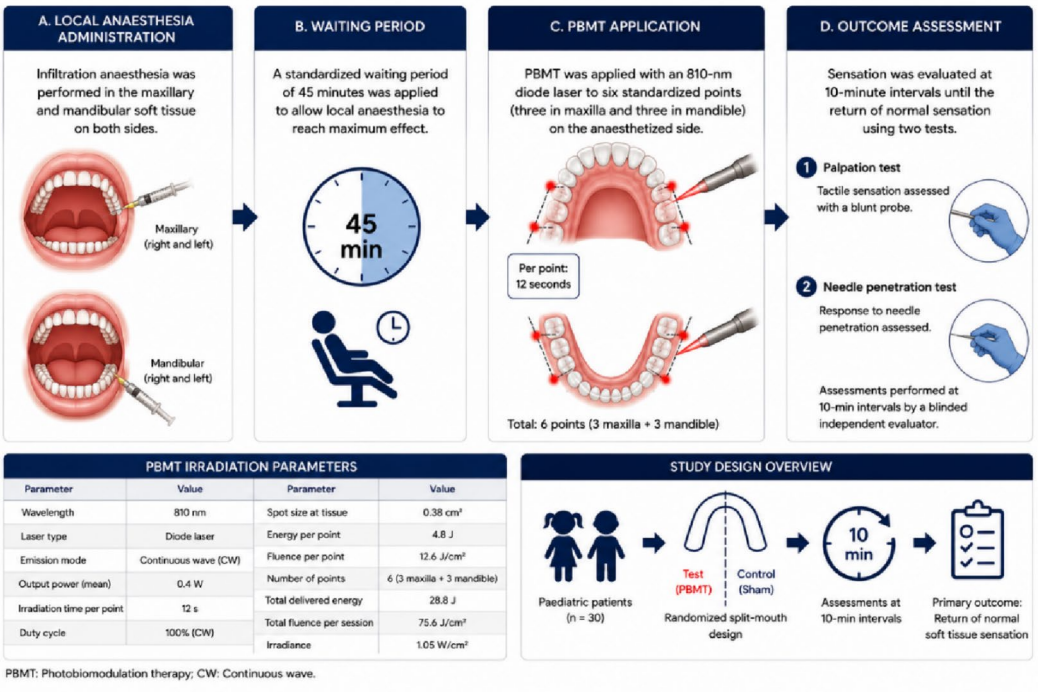


Fig. 2. Schematic overview of the study design, PBMT application protocol, and outcome assessment.

Overall (*N* = 40)	
Gender	
Male	16 (40.0%)
Female	24 (60.0%)
Age	
Mean (SD)	7.3 (1.4)
Range	4.0–9.0
Age range	
>7 years	19 (47.5%)
≤7 years	21 (52.5%)
Frankl behaviour rating score	
3	21 (52.5%)
4	19 (47.5%)

Table 2. Demographic characteristics of participants.

Outcome	PBMT group	Sham group	*p*-value
Mean recovery time (min), mean ± SD	163.0 ± 53.0	206.9 ± 83.8	< 0.001
Median recovery time (min)	157.5	195.0	-
Return of normal soft tissue sensation at 10 min, n (%)	7 (17.5%)	0 (0%)	NC
Return of normal soft tissue sensation at 20 min, n (%)	18 (45%)	4 (10%)	0.002
Return of normal soft tissue sensation at 30 min, n (%)	29 (72.5%)	14 (35%)	0.035

Table 3. Return of normal soft tissue sensation in PBMT and sham groups.

Discussion

The present randomized split-mouth clinical trial demonstrated that PBMT using an 810-nm diode laser significantly reduced the duration of soft tissue local anaesthesia in paediatric patients. Laser-treated sites exhibited faster recovery of normal sensation and a lower incidence of post-operative soft tissue trauma

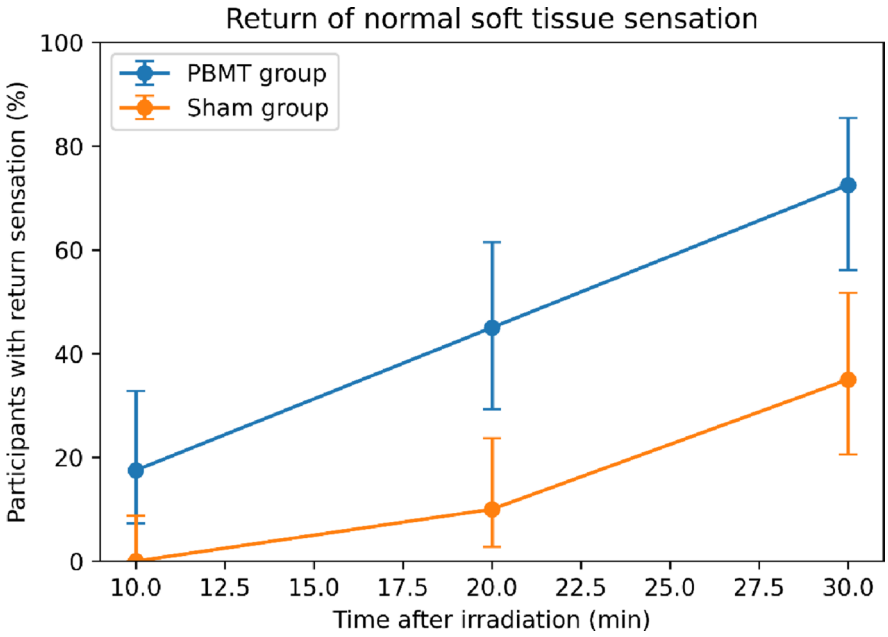


Fig. 3. Percentages of participants with return of normal soft tissue sensation at 10, 20, and 30 min after PBMT application in the PBMT and sham groups. Vertical error bars indicate 95% confidence intervals.

	PBMT group	Sham group	p-value
Gender			
Male	170.63 (60.77)	210.63 (78.48)	0.065[1]
Female	157.92 (47.82)	204.38 (88.75)	0.008[1]
p-value	0.465[2]	0.821[2]	
Age range			
>7	177.89 (61.18)	220.53 (98.6)	0.039[1]
≤7	149.52 (41.32)	194.52 (67.9)	0.013[1]
p-value	0.091[2]	0.334[2]	
Dental Arches			
Mandibula	165.5 (38.62)	207.25 (92.22)	0.045[1]
Maxilla	160.5 (65.27)	206.5 (76.9)	0.013[1]
p-value	0.77[2]	0.978[2]	

Table 4. Return of normal soft tissue sensation according to dental arch in PBMT and sham groups. [1]Paired samples t-test, [2]Student's t-test

compared with sham-treated sites. These findings support the hypothesis that PBMT can effectively accelerate anaesthetic reversal in children and are consistent with previously reported data.

The biological mechanisms underlying PBMT-mediated anaesthetic reversal are thought to involve enhanced microcirculation and endothelium-dependent vasodilation[6,9,11,21]. Near-infrared laser irradiation has been shown to stimulate nitric oxide release, increase mitochondrial activity, and improve local blood flow, thereby facilitating faster clearance of local anaesthetic agents from soft tissues[18,19,21]. The wavelength used in the present study (810 nm) provides adequate tissue penetration while maintaining a favourable safety profile, making it suitable for paediatric applications[6,7,9,15,22,23].

Previous clinical studies have primarily focused on the mandibular region when evaluating PBMT for anaesthetic reversal[6,9]. Seraj et al. reported a significant reduction in mandibular soft tissue anaesthesia duration following application of an 810-nm diode laser in paediatric patients[9]. Similarly, Annu et al. demonstrated accelerated recovery of soft tissue sensation using both 660-nm and 810-nm diode laser wavelengths[6]. In addition, Olszewska et al. investigated PBMT-mediated reversal of local anaesthesia in the maxillary arch and demonstrated earlier return of normal sensation following irradiation with 635-nm and 808-nm diode lasers compared with sham treatment[20]. Their findings also suggested a potential reduction in self-inflicted soft tissue trauma associated with earlier sensory recovery[20].

The findings of the present study are consistent with previous reports and further extend the existing evidence by demonstrating that PBMT may be effective in both maxillary and mandibular arches within the

same paediatric population. Unlike earlier investigations that evaluated a single anatomical region, the present randomized split-mouth clinical trial included both arches using a standardized PBMT protocol and comparable outcome assessment methods. Subgroup analyses demonstrated similar recovery trends in maxillary and mandibular procedures, suggesting that the beneficial effects of PBMT may not be limited to a specific dental arch. From a clinical perspective, this finding is important because both maxillary and mandibular infiltration anaesthesia are frequently performed in paediatric dental practice, and prolonged soft tissue numbness may negatively affect eating, speaking, and postoperative comfort regardless of the treated region.

To the best of our knowledge, this is the first randomized clinical trial to evaluate PBMT-mediated reversal of soft tissue anaesthesia in both dental arches using a split-mouth design. Nevertheless, the subgroup findings should be interpreted cautiously, as the study was not specifically powered to detect arch-related differences and the sample size remained relatively limited. Khosraviani et al. highlighted the therapeutic benefits of laser applications in the management of paediatric oral soft tissue conditions[14]. In the present study, PBMT not only shortened the anaesthesia time but also significantly reduced the frequency of soft tissue trauma cases, thus having a positive indirect effect on soft tissue. This clinical outcome is particularly relevant in paediatric dentistry, where accidental lip or cheek biting remains a common complication of prolonged soft tissue numbness. Compared with pharmacological reversal agents such as phentolamine mesylate[5,8], PBMT offers a non-invasive, drug-free alternative that may be more acceptable to both children and caregivers, while avoiding potential systemic effects. The shorter recovery time observed in the PBMT group may be associated with the biological effects of photobiomodulation therapy on tissue repair and inflammatory modulation. Farhad et al. demonstrated that 810-nm diode laser irradiation increased the expression of mediators associated with tissue healing and remodelling while modulating inflammatory response pathways[15].

Subgroup analyses revealed that PBMT was effective across age groups and dental arches, with a significant gender-related difference observed only among female participants. Although the clinical relevance of this finding remains unclear, it may be related to sex-based differences in pain perception, vascular responsiveness, or tissue sensitivity. Further studies are required to elucidate the underlying mechanisms of this observation.

Several limitations should be considered when interpreting the findings of the present study. First, although the split-mouth randomized design reduced interindividual variability, the possibility of carry-over effects and behavioural influences between appointments cannot be completely excluded. Second, the sample size was relatively limited, and subgroup analyses according to dental arch should therefore be interpreted cautiously. Third, recovery of soft tissue sensation was assessed clinically using subjective patient responses, which may have introduced response variability, particularly in younger children. In addition, the findings are limited to the specific PBMT parameters, wavelength, and irradiation protocol used in this study and may not be generalizable to other laser settings or clinical conditions. Another limitation is that the study evaluated short-term clinical outcomes only; therefore, long-term effects of PBMT on sensory recovery and patient comfort could not be assessed. Finally, although both maxillary and mandibular procedures were included, the study was not specifically powered to detect differences between dental arches.

Conclusion

Within the limitations of this study, PBMT using an 810-nm diode laser appears to be a safe, non-invasive, and clinically useful adjunct for accelerating recovery of normal soft tissue sensation following local anaesthesia in paediatric dental patients. PBMT was associated with shorter recovery times and reduced soft tissue trauma incidence in both maxillary and mandibular procedures. Further randomized clinical studies with larger sample sizes and standardized PBMT protocols are required to confirm these findings and optimize clinical application parameters.

Data availability

Data is provided within the manuscript or supplementary information files.

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Author contributions

AK, ZG and EB conceived the ideas. AK, EK and HY collected the data. AK analysed the data. AK and HY led the writing. All authors reviewed the manuscript. As corresponding author A.K then formatted the paper and ensured it met all relevant presubmission guidelines.

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Declarations

Competing interests

The authors declare no competing interests.

Human ethics and constent to participate

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. The study was approved by the Invasive Clinical Research Ethics Committee of Recep Tayyip Erdogan University (Decision no: 2022/06).

Additional information

Correspondence and requests for materials should be addressed to A.K.

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